Sky Subtraction

I. Sky Subtraction

In the case of both infrared and visual imaging, the emission from the skylight will be super-positioned with signal from the object itself, and therefore needs to be subtracted. For the infrared, the sky is more dominant than in the visual (see **??**), but the approach to the subtraction from a data processing point of view if somewhat similar. In this chapter, three different sky subtraction approaches are described as implemented in the pipeline. All of these are dictated by the observational technique used for the observations. These observational techniques are also described briefly.

A. Median sky subtraction

In the simplest case, where the object is observed with a single frame, or with several frames without using dithering (see the following chapter for explanation of dithering), the simple approach of a **median sky subtraction** is used. In this case, the median pixel count value is calculated in the reduced frame, and simply subtracted from every pixel on the detector. This method works under two bold assumptions. Firstly, it assumes that sky is filling most of the frame i.e. that the median value will in fact be a pixel holding a median value of the sky. Secondly, it assumes that the sky is uniform throughout the whole detector as only a constant value is subtracted. These assumptions are in practise often not met, and therefore **this approach is more a fallback** for frames that were observed without dithering or off-beam sky observations as described in the remainder of these chapter. In practise, either dithering or off-beam sky observations should be performed for observations where correct sky subtraction is needed.

B. Dithered sky subtraction

For sources that are not significantly extended, an effective observing strategy to estimate the sky is **dithering**. Dithering is easiest explained by visualisation as shown in figure 1 below.

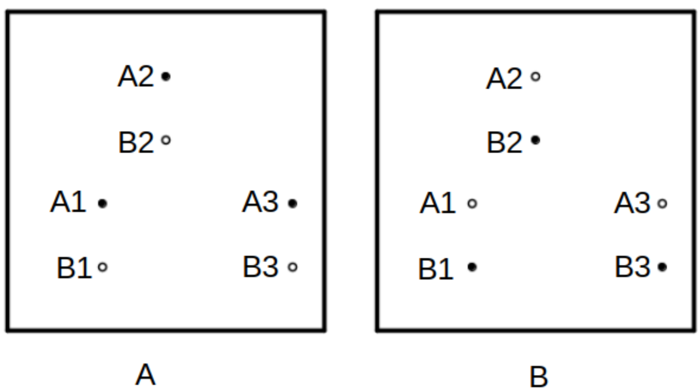

Fig. 1: Figure illustrating the principle of dithering using two frames A and B, with the telescope off-set being applied slightly downwards from frame A to frame B. The filled circles represent objects (1,2,3), while the empty circles represent the corresponding positions of these objects in both frames. By applying a telescope off-set, the sky at the pixels where the objects are placed in frame A is recorded on frame B, and vice-versa.

In observations with dithering, multiple images are taken of the same object, while off-setting the telescope by around tens of arcseconds, making the object shift slightly on the detector. This way, when the telescope gets off-set from frame A to frame B, the sky at the detector position of the object in frame A gets recorded in frame B and vice-versa (see figure 1). The sky can then get subtracted from frame A by subtracting B from A, and identically A from B for the B frame. Ideally, more than two frames should be taken (all with different telescope off-sets), as this allows to construct a **sky frame** by median combining all dithered observations. For a stack of dithered frames, for a given detector pixel, the assumption here is that the majority of the frames will have only sky on that pixel, and only few will have an object. By taking the median for every pixel of the whole stack, all median pixels should ideally hold only the counts corresponding to the sky. An example of this is shown in figure 2.

Even when the separate exposures are conducted consequentially, the observed field will gradually either gain or loose **airmass** (amount of atmosphere between the telescope and the object) during the sequence. This means that the sky brightness will vary slightly between the different exposures of the same field. Therefore, **scaling** is necessary both when enforcing simple A-B subtraction and when using median combination to construct a master sky frame. In the pipeline, the scaling factor is calculated as the quotient of medians between frames. For A-B subtraction, B is scaled to A. For master sky frame subtraction, all exposures all first (temporarily) scaled to the first observation (arbitrary choice), before median combining the frames, and then the master sky frame gets scaled to every frame it is applied to.

Dithering has other benefits besides possibility for efficient sky estimation. Shifting the object on the detector enforces a larger amount of different pixels to record the same object signal. This can be beneficial for several reasons. Firstly, if the object falls on defect pixels in frame A, moving the object in frame B will likely place the object on healthy ones. Furthermore, even when the images are flat fielded (see **??**), the calibration for pixel-to-pixel sensitivity will never be perfect, and enforcing different detector areas to record the same object will statistically minimize the impact of difference between pixel sensitivity when the dithered images get stacked (see **??**).

C. Off-beam sky subtraction

For observations of extended objects, dithering for sky estimation can be problematic, as the object might be too extended with respect to the detector size to allow fully shifting the object to a set of completely new set of pixels. This is illustrated in figure 3. In such cases, in order to estimate the sky, the telescope can be off-set to a nearby field with as few as possible sources that are non-extended, completely removing the object from the detector, and recording only the sky. In practise, this sequence (object - sky) is looped several times. The individual object exposures and the sky exposures should still be dithered for reasons described in the preceding chapter. This technique is often referred to as **off-beam sky observing**. At the NOT, this is only used for NOTcam observations, but is also intended to be one of the observing modes that NTE offers.

The off-beam sky subtraction in the pipeline is implemented almost identically to the dithered sky subtraction (see preceding chapter), the only difference is that the master sky frame is now constructed from the off-beam exposures, and not the object exposures themselves. Another difference is that a master sky frame can now essentially

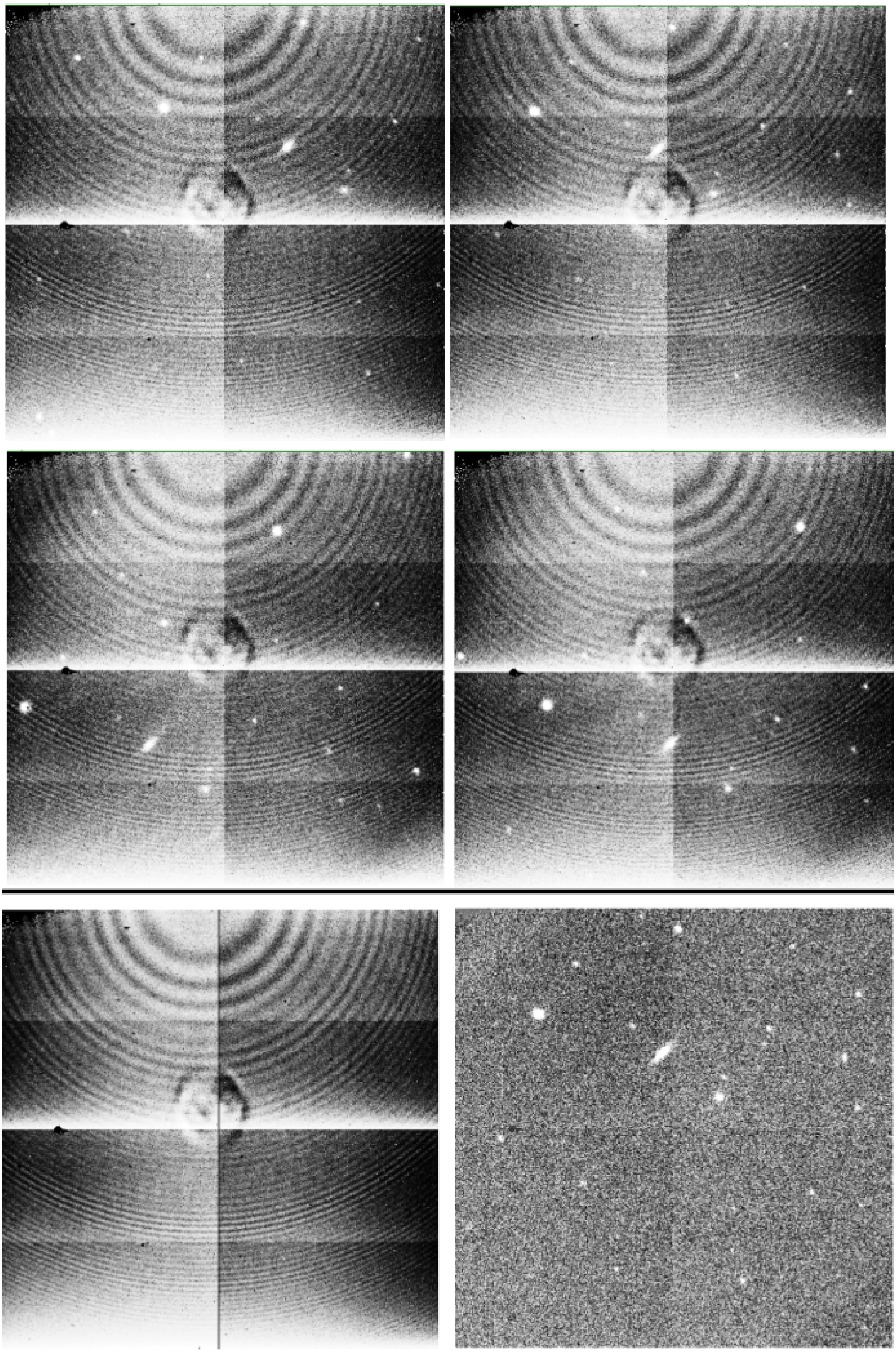

Fig. 2: Figure illustrating sky subtraction using dithered images. The top four images are reduced NOTcam images in H band taken with different telescope off-sets. The lower left image is a median stack of all exposures i.e. the master sky frame (of in all eight dithers - all not shown in this figure). The lower right is the difference between the upper right frame and the sky frame.

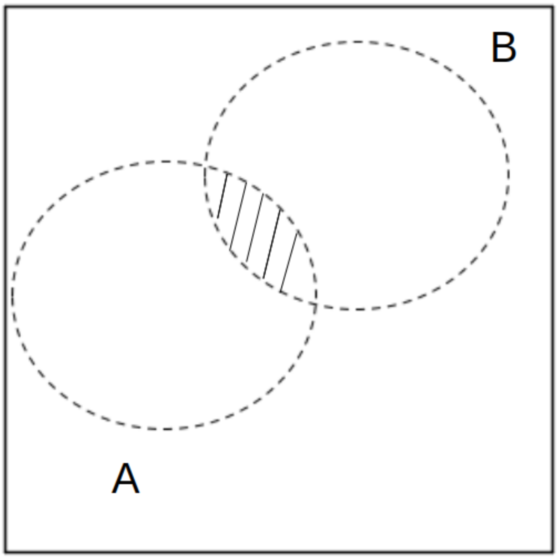

Fig. 3: Figure illustrating how dithering might not be sufficient when estimating sky for an extended object. The dashed ellipses represent the object placement for dithers A and B (see figure 1 for reference). The marked intersection between the two object positions show a detector region where the sky won't be recorded for any of the frames.

be constructed by one off-beam frame (even though not recommended), while with dithering, a minimum of three frames is needed.

The downside of off-beam sky estimations, compared to estimating the sky with dithered science frames, is that it costs extra telescope time. The time used for off-beam observing is equal to the time used for observing the science object, essentially doubling the cost of the observation. However, for extended sources, this might be necessary especially in the infrared, where the radiation from the sky is dominating over the actual radiation of the science object.

D. Determining what method to use for sky subtraction

As the pipeline is intended to run fully automatic, robust logic is needed to determine what method for sky subtraction should be used for an observation. The logic implemented in the code is shown in figure 4.

Whether sky subtraction is at all wanted for an observation is not trivial to determine. In some cases, sky subtraction might not be optimal, for example if the field is extremely crowded with no off-beam observations available, or simply the case where sky subtraction is not essential for the particular science case. For this reason, the pipeline saves copies of the frames both with and without the sky subtraction.

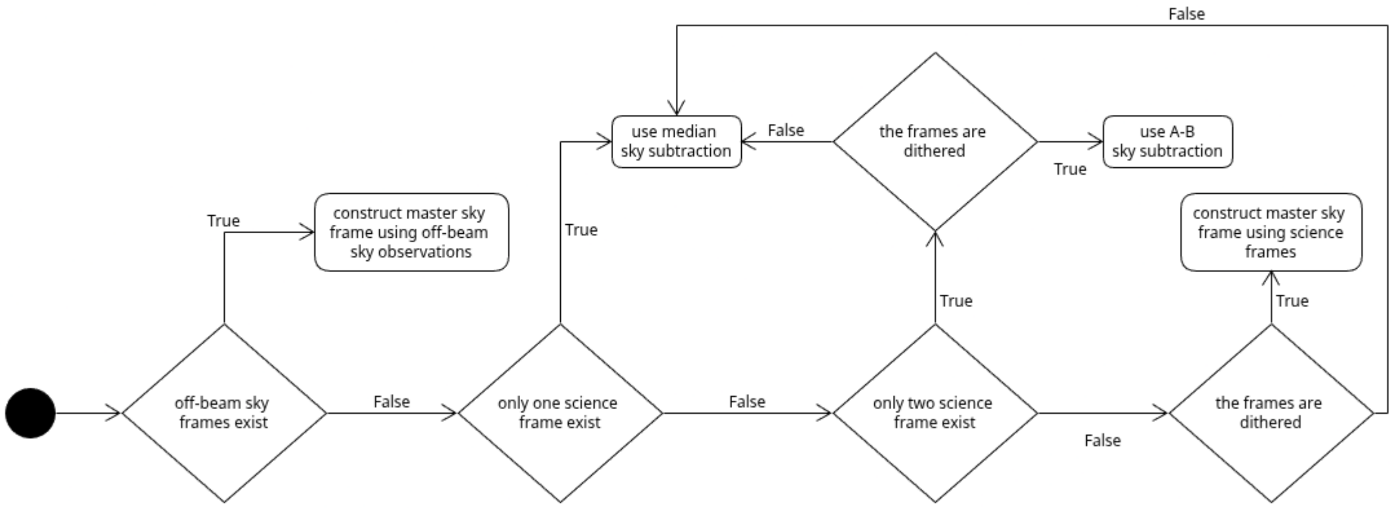

Fig. 4: Figure illustrating the decision logic implemented into the pipeline regarding what method should be used for the sky subtraction. The figure is intended to be followed from left to right.